HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 19 Name: _____ Marks: / 70

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The solution of $4y+1<5y-3\leq 8y-9$ is

A. $y>-4$.

B. $y\geq -2$.

C. $y\geq 2$.

D. $y>4$.

(DSE-2025 PII Q07)

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2. The perimeter of the sector OMN is 12π cm, where O is the centre of the sector OMN. If the radius of the sector OMN is 3π cm, which of the following are true?

I. The area of the sector OMN is $9\pi^2$ cm^2.

II. The perimeter of $\triangle OMN$ is less than 35 cm.

III. The angle of the sector OMN is greater than $100°$.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

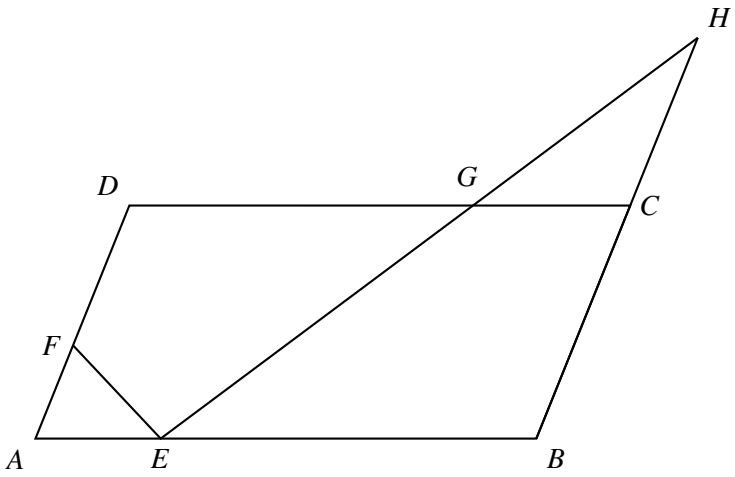
(DSE-2025 PII Q15)

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3. In the figure, $ABCD$ is a parallelogram. Let E, F and G be points lying on AB, AD and CD respectively such that $BE = 3AE$, $2DF = 3AF$ and $DG = 2CG$. It is given that BC produced and EG produced meet at the point H. If the area of $\triangle CGH$ is $16\,\text{cm}^2$, then the area of the quadrilateral $DFEG$ is



A. $46\,\text{cm}^2$.

B. $49\,\text{cm}^2$.

C. $105\,\text{cm}^2$.

D. $115\,\text{cm}^2$.

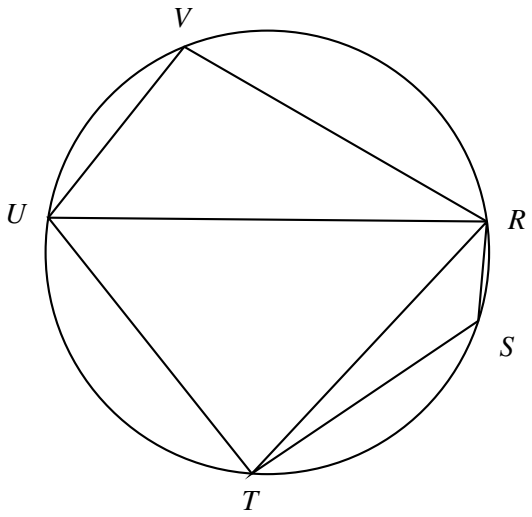
(DSE-2025 PII Q17)

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4. In the figure, $RSTUV$ is a circle. It is given that $RT \,//\, VU$ and RT is the angle bisector of $\angle SRU$. If $\angle RTS = \angle URV = 33°$, then $\angle RUT =$



A. 36°.

B. 49°.

C. 65°.

D. 71°.

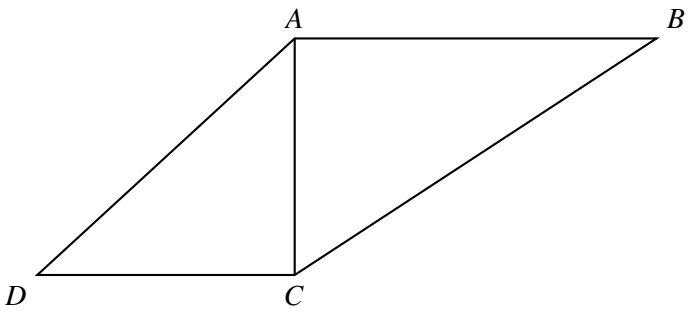
(DSE-2025 PII Q22)

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5. The figure shows the quadrilateral $ABCD$. If $\angle ABC + \angle ADC = \angle ACD = \angle BAC = 90°$, which of the following must be true?



A. $\tan\angle ACB = \dfrac{AB}{AD}$

B. $\tan\angle ACB = \dfrac{AB}{CD}$

C. $\tan\angle ADC = \dfrac{BC}{AD}$

D. $\tan\angle ADC = \dfrac{BC}{CD}$

(DSE-2025 PII Q23)

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6. Denote the origin by O. The coordinates of the point A are $(a, 2a)$, where $a > 0$. Let P be a moving point in the rectangular coordinate plane such that $AP = OA$. The locus of P is a

A. point.

B. circle.

C. rhombus.

D. straight line.

(DSE-2025 PII Q25)

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7. The straight lines L_1 and L_2 are perpendicular to each other. The equations of L_1 and L_2 are $3x+4y-20=0$ and $mx+ny-20=0$ respectively, where m and n are constants. Suppose that L_1 and L_2 cut the x-axis at the points A and B respectively. Let C be the point of intersection of L_1 and L_2. It is given that C lies above the x-axis. If the area of $\triangle ABC$ is 6, find n.

A. -12

B. -9

C. 9

D. 12

(DSE-2025 PII Q26)

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8. The coordinates of the centre of the circle C are $(7,-5)$. If C cuts the x-axis at the points P and Q such that $PQ = 24$, then the equation of C is

A. $x^2+y^2-14x+10y-95=0$.

B. $x^2+y^2+14x-10y-95=0$.

C. $x^2+y^2-14x+10y-119=0$.

D. $x^2+y^2+14x-10y-119=0$.

(DSE-2025 PII Q27)

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9. A box contains six cards numbered 2, 2, 2, 3, 4 and 5 respectively. In a game, a number is randomly drawn from the box and a certain number of tokens will be got according to the following table:

Number drawn	2	3	4	5
Number of tokens got	10	15	25	50

Find the expected number of tokens got in the game.

A. 3

B. 4

C. 20

D. 25

(DSE-2025 PII Q28)

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10. $3E000000000000_{16} =$

A. $2^{16}+2^{15}+2^{14}+2^{13}+2^{12}$.

B. $2^{17}+2^{16}+2^{15}+2^{14}+2^{13}$.

C. $2^{52}+2^{51}+2^{50}+2^{49}+2^{48}$.

D. $2^{53}+2^{52}+2^{51}+2^{50}+2^{49}$.

(DSE-2025 PII Q31)

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11. It is given that $\log_5 y$ is a linear function of $\log_{25} x$. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 12 and 2 respectively. If $y = mx^n$, then $n =$

A. -12.

B. -4.

C. -3.

D. -1.

(DSE-2025 PII Q33)

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12. On the same rectangular coordinate system, the graph of $y = \log_a x$ cuts the graph of $y = a^x$ and the x-axis at the points P and Q respectively, where a is a positive constant. Denote the origin by O. Which of the following are true?

I. $a < 1$

II. $OQ > a$

III. $\angle POQ = 45°$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q34)

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13. Consider the following system of inequalities:

$$\begin{cases} x \leq 11 \\ 4x + 5y - 19 \geq 0 \\ 7x - 6y + 11 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D, then the greatest value of $8x - 6y + 11$ is

A. 1.

B. 11.

C. 15.

D. 129.

(DSE-2025 PII Q36)

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14. Let p, q and r be non-zero real numbers. If p, q, r is an arithmetic sequence, which of the following must be true?

I. 3^p, 3^q, 3^r is a geometric sequence.

II. $\frac{5}{p}$, $\frac{5}{q}$, $\frac{5}{r}$ is a geometric sequence.

III. $p-q$, $q-r$, $r-p$ is an arithmetic sequence.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2025 PII Q37)

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15. For $90° < \theta < 270°$, how many roots does the equation $\tan^3\theta = 2\tan\theta$ have?

A. 2

B. 3

C. 4

D. 5

(DSE-2025 PII Q39)

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16. Denote the origin by O. The coordinates of the point U are $(20, 0)$. Let V be a point lying on the positive y-axis such that the x-coordinate of the in-centre of $\triangle OUV$ is 6. Find the area of $\triangle OUV$.

A. 70

B. 87

C. 210

D. 250

(DSE-2025 PII Q41)

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17. Let $f(x) = 2x^3 + hx^2 + kx + 15$, where h and k are constants. When $f(x)$ is divided by $x+2$, the remainder is -45. It is given that $f(x)$ is divisible by $2x-5$.

(a) Find h and k.

(b) Someone claims that all the roots of the equation $f(x) = 0$ are rational numbers. Do you agree? Explain your answer.

(6 marks) (DSE-2025 PI Q10)

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18. The straight line L cuts the positive x-axis and the positive y-axis at the points M and N respectively. Let P be a moving point in the rectangular coordinate plane such that $MP = NP$. Denote the locus of P by Γ. It is given that the equation of Γ is $3x - 2y - 30 = 0$.

(a) Describe the geometric relationship between Γ and MN.

(b) Find the equation of L.

(c) Suppose that Γ cuts L and the x-axis at the points Q and R respectively. Find the area of the quadrilateral $ONQR$, where O is the origin.

(7 marks) (DSE-2025 PI Q13)

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19. It is given that $\begin{cases} \log_3 x + \log_3 y = 9 \\ \log_x 81 - \log_y 9 = 1 \end{cases}$, where $0 < x < y$.

(a) Let $u = \log_3 y$. Prove that $u^2 - 3u - 18 = 0$.

(b) Find x.

(4 marks) (DSE-2025 PI Q16)

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20. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where k is a positive constant. Denote the vertex of the graph of $y = g(x)$ by R.

(a) Using the method of completing the square, express the coordinates of R in terms of k.

(b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by S. Let T be a point such that the orthocentre of $\triangle RST$ is the origin.

(i) Express the coordinates of T in terms of k.

(ii) The coordinates of the point U are $(-5, 5)$. Let V be a point lying on RS such that TV is perpendicular to RS. It is given that S, T, U and V are concyclic. Find k.

(9 marks) (DSE-2025 PI Q18)

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21. The equation of the straight line L is $4x-3y+83=0$. The equation of the circle C is $x^2+y^2+ax-2y+b=0$, where a and b are positive constants. It is given that C passes through the point $(-10,9)$ and L is a tangent to C.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find a and b.

(b) The coordinates of the point P are $(-20,1)$. Let Q be a point lying below L such that PQ is tangent to C and $PQ=25$. Denote the centre of C by I.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between I and Q.

(iii) Let R be a point lying on L such that C is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by r. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

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